

Case 4: $n > m \geq 3$. Again, since $x^{1/x}$ decreases, $m^{1/m} > n^{1/n}$ or $m^n > n^m$ or $m^n(\log n) > n^m(\log m)$ (a fortiori) or $n^{(m^n)} > m^{(n^m)}$.

§7.2. The infinite exponential $x^{x^{x^{\dots}}}$

We have already mentioned that given ξ , the corresponding η satisfying the equation $\xi^\eta = \eta^\xi$, is not readily available, even after the parameterization $x = s^{1/(s-1)}$, $y = s^{s/(s-1)}$, where $s \in \mathbb{R}^+$. For example, if say $\xi = \sqrt{2}$ (so that $1 < \xi < 2$), trying to get s from the equation $\sqrt{2} = s^{1/(s-1)}$ is not possible and without this s , the corresponding $\eta = s^{s/(s-1)}$ is not available. Thus we really have to have something even better than the above parameterization. This is exactly where the beautiful “function”

$y = h(x) = x^{x^{x^{\dots}}}$, i.e., y equal to the limit of the sequence $x, x^x, x^{x^x}, \dots$, (*)

comes into the picture. Before we consider the connection of this function with that of $f(x, y) = x^y - y^x$, we must determine the values of x for which $h(x)$ is meaningful. Thus, for example, if $x = 2$, the sequence $2, 2^2, 2^{2^2}, \dots$ is obviously divergent and it would seem that for $x > 1$, the sequence would diverge. However, it surprises most to learn that there are values of x greater than 1, for which $h(x)$ converges. As it turns out:

Theorem 7.2.1. *The infinite exponential, as defined above, converges for all values of x in the range $[e^{-e}, e^{1/e}]$, i.e., for all x satisfying*

$$.06598803585 \dots = e^{-e} \leq x \leq e^{1/e} = 1.444667861 \dots$$

We first give a graphical motivation of this result (see D.F. Barrow [8] or M.C. Mitchelmore [68]).

Graphical proof (motivation) of Theorem 7.2.1. Let $\zeta_1 = a$, $\zeta_2 = a^a$, $\zeta_3 = a^{a^a} = a^{\zeta_2}$, $\dots$, $\zeta_n = a^{\zeta_{n-1}}$ and ask for what values of a , the limit, as n tends to infinity, of ζ_n exists. The trick is to consider the graph Γ of the function $y = a^x$, which, in spite of having a more or less uniform shape, varies a great deal, for different values of a , in its orientation and its relative position to the graph G of the line $y = x$, and it is this fact that will be seen to account for convergence or divergence of the sequence $\{\zeta_n\}$ for various values of a . We need to split different cases:

Case 1: $1 < a < e^{1/e}$. To fix our ideas, let us take $a = \sqrt{2} = 1.4142 \dots$. The graph Γ and G of $y = (\sqrt{2})^x$ and $y = x$ respectively are shown in Figure 7.6. We see that the set $\Gamma \cap G = \{(2, 2), (4, 4)\}$ and the points $Q_1 = (\sqrt{2}, \sqrt{2}^{\sqrt{2}})$, $Q_2 = (\sqrt{2}^{\sqrt{2}}, \sqrt{2}^{\sqrt{2}^{\sqrt{2}}})$, $\dots$ can be successively got by chasing the zigzag shown in Figure 7.6, starting with $a = \sqrt{2}$ on the x -axis and going up to Q_1 on Γ ,

then to P_1 on G , onto Q_2 on Γ , and so on. It is clear from the graph that $Q_i \rightarrow (2, 2)$, showing that $\sqrt{2}^{\sqrt{2}^{\cdot^{\cdot^{\cdot}}}} \rightarrow 2$.

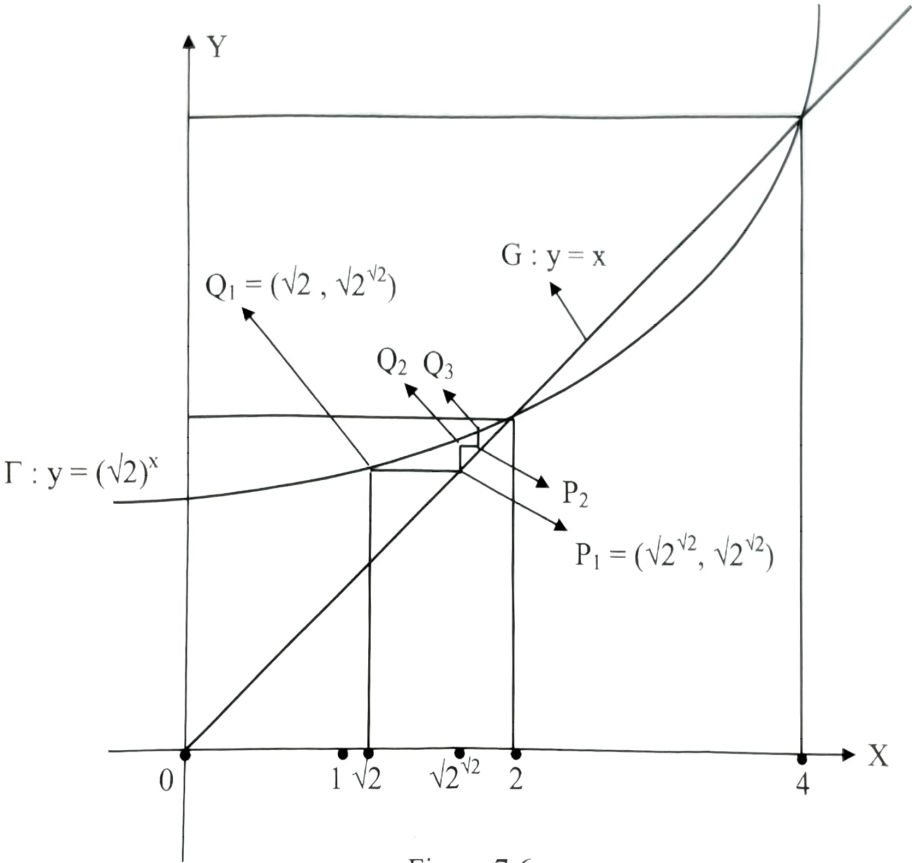


Figure 7.6

However, there is a second point of intersection of Γ and G , viz. $(4, 4)$, but clearly the process stops at the point with smaller coordinates (we shall prove the existence of this limit analytically as well; however, the graphical “proof” is far more revealing, illuminating and instructive).

Case 2: $a = e^{1/e}$. This zigzag process described above fails if Γ and G do not intersect in real points. As a increases, a time will come when G is tangent to Γ . The critical value, when this happens, is $a = e^{1/e}$. So let us draw the graph Γ of $y = (e^{1/e})^x$ (see Figure 7.7).

The point $(e, e) \in \Gamma$. Let us determine the tangent to Γ at (e, e) . We have $y = e^{x/e}$ and so $\log y = x/e$, giving $dy/dx = y/e = e^{x/e}/e$. At the point (e, e) , the value of dy/dx is equal to $e/e = 1$. The equation of the required tangent is therefore $y = x + c$. Since this goes through (e, e) , $c = 1$ and so the tangent is the line $y = x$, as required.

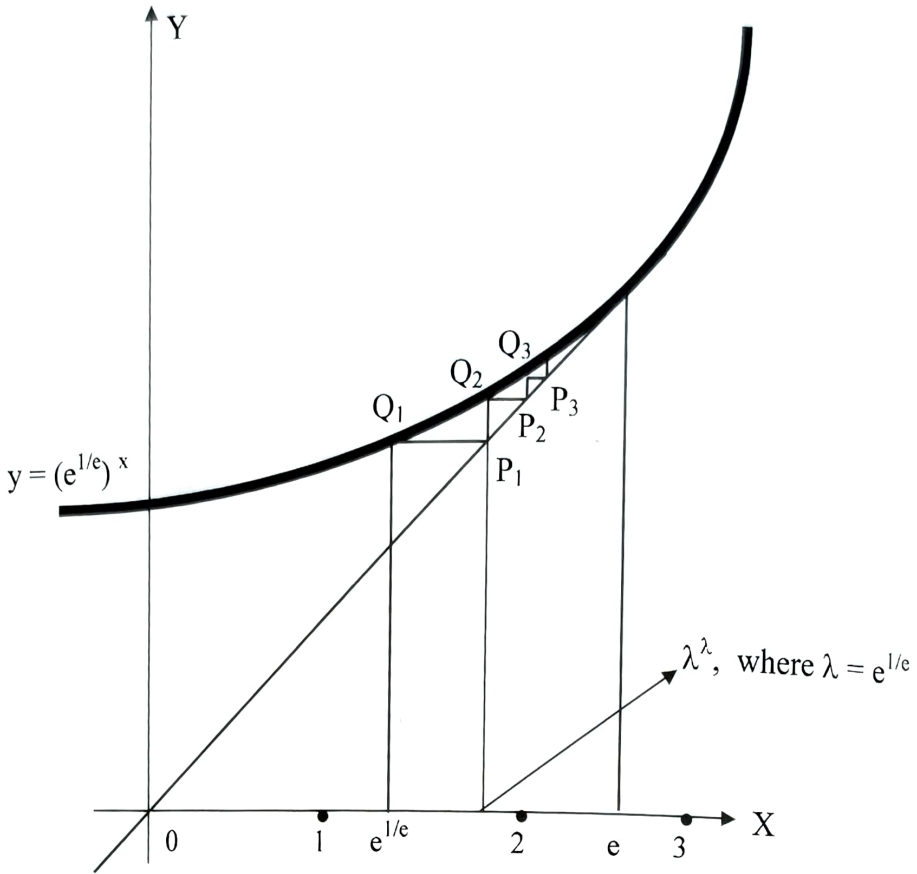


Figure 7.7

Now, taking $x = e^{1/e}$ on the x -axis, we see that the zigzag of points $Q_1, Q_2, \dots \rightarrow (e, e)$, in particular,

$$(e^{1/e})(e^{1/e})^{\cdot\cdot\cdot} \rightarrow e.$$

Case 3: $a > e^{1/e}$. The curve $\Gamma : y = a^x$ has graph shown in Figure 7.8, and it is obvious that for such values of a , the infinite exponential $a^{a^{\cdot\cdot\cdot}}$ diverges.

Case 4: $1 < a < e^{1/e}$. Here, exactly as for the case $a = \sqrt{2}$ in case 1, the sequence

$$a^{a^{\cdot\cdot\cdot}}$$

converges. There will, of course, be 2 points of intersection of Γ and G , since $a^x \rightarrow \infty$ faster than x (since $a > 1$) and at $x = 0, y = 1$, so Γ must cross G twice. The graph of various a , $1 \leq a \leq e^{1/e}$ are shown in Figure 7.9 and $a^{a^{\cdot\cdot\cdot}}$ converges to the smaller of the x -coordinates of the two points of intersection.

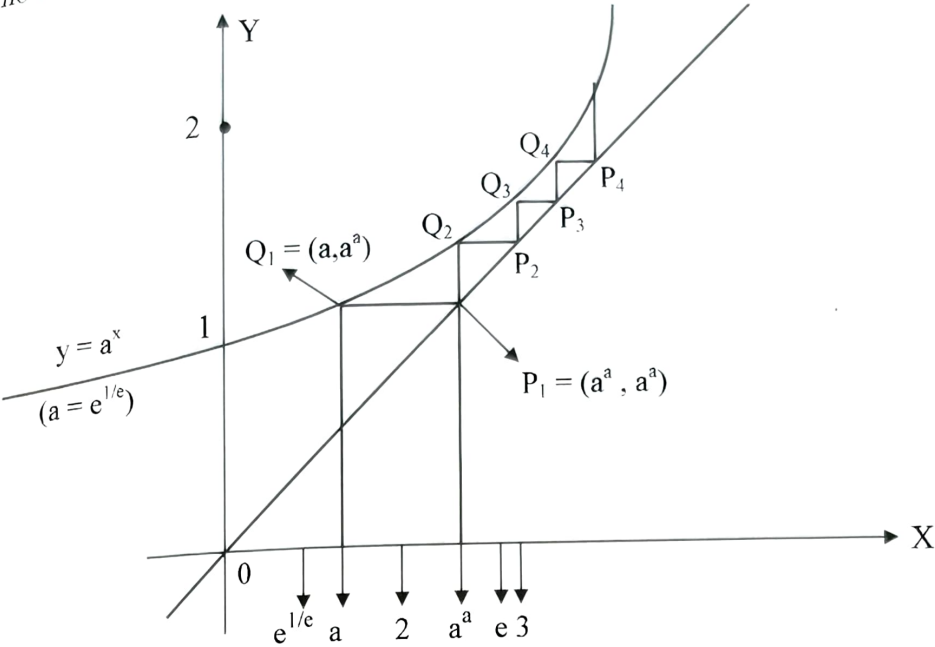


Figure 7.8

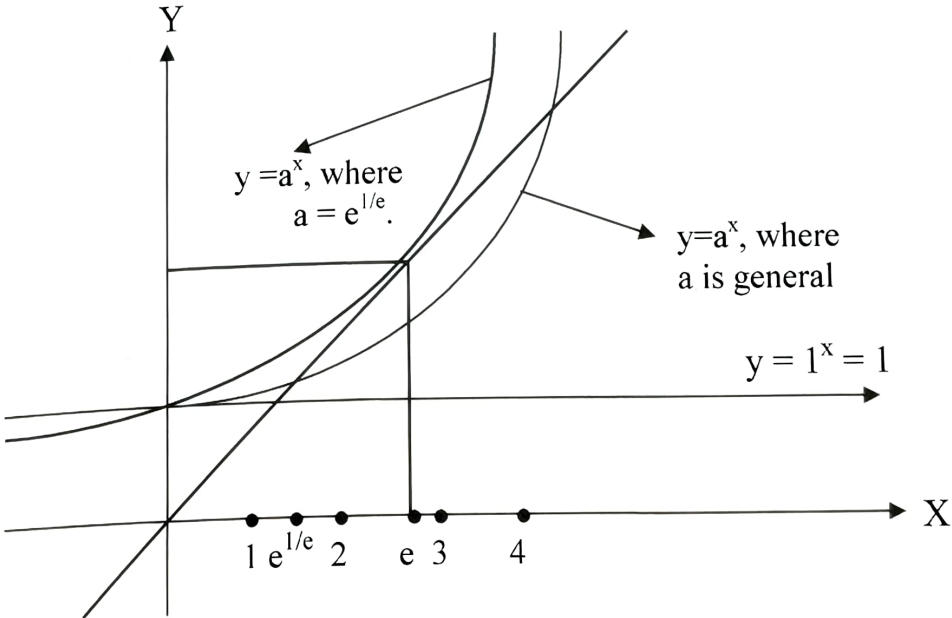


Figure 7.9

For various $a < 1$, the graphs of $y = a^x$ behave differently, since, as $x \rightarrow \infty$, $a^x \rightarrow 0$ and as $x \rightarrow -\infty$, $a^x \rightarrow \infty$ and so the graphs Γ have shape shown in Figure 7.10.

As it turns out, we need to consider the graph

$$\Gamma^* : x = a^y$$

as well, which is the mirror image of $\Gamma : y = a^x$ in the line $G : y = x$ and we further need to determine the points $\Gamma \cap \Gamma^*$ in this case (i.e., when $0 < a < 1$),

i.e., all the real solutions of the pair of equations

$$(i) y = a^x \quad \text{and} \quad (ii) x = a^y \quad (*)$$

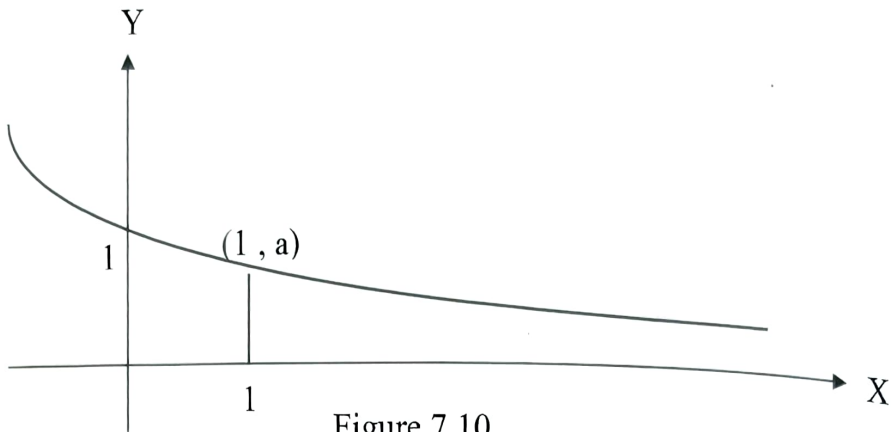


Figure 7.10

Now one solution of $(*)$ is $x = y = \zeta$, say. Let P be the point (ζ, ζ) . If the absolute value of the gradient of (i) at P is greater than 1, then clearly the graph of (i) will drop below the graph of (ii) as x moves to the right of P and further since it stays above the x -axis, it follows that the two graphs must cross each other once more, at Q , say. By symmetry about G , there is then a third point R of intersection of Γ and Γ^* (see Figure 7.11).

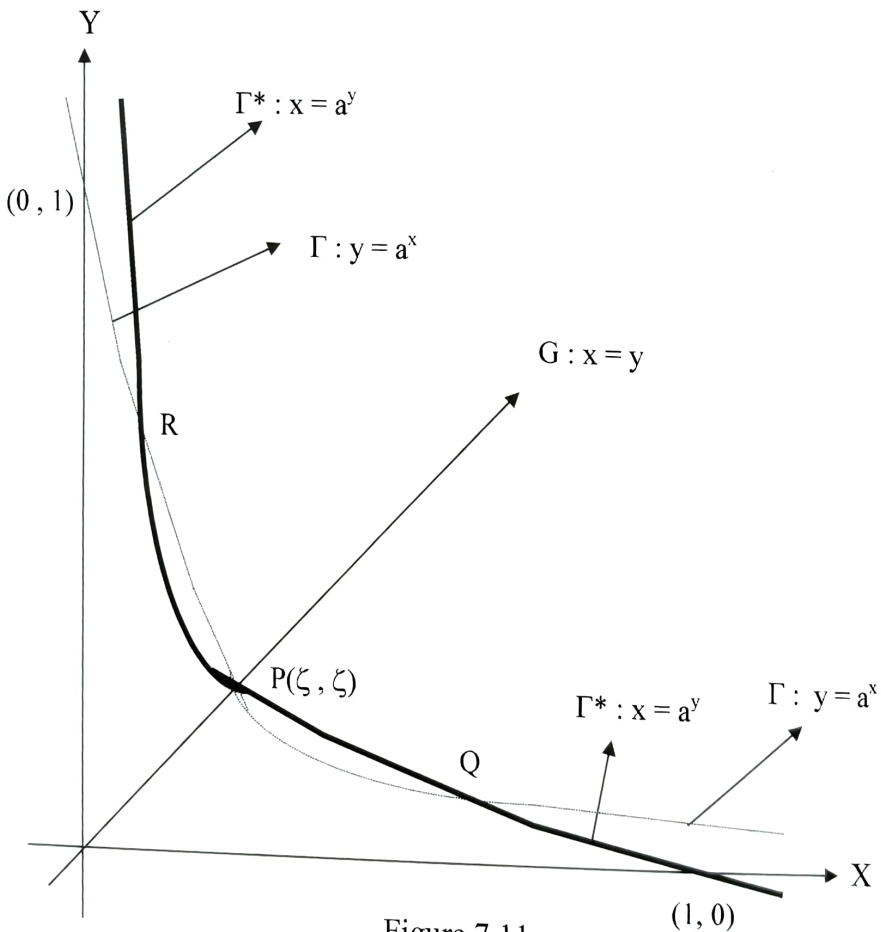


Figure 7.11

The problem now is to find when (i.e., for what values of a) the graph Γ has absolute value of the gradient greater than 1 at P . We shall see that the critical value is at $a = e^{-e}$. For then, our Γ is $y = (e^{-e})^x = e^{-ex}$ and so $dy/dx = e^{-ex} \cdot \log(e^{-e}) = -e \cdot e^{-ex}$. To get P , we solve $y = e^{-ex}$ with $y = x$, giving $e^{-ex} = x$. Since there is only one real solution (because Γ is decreasing) and $x = 1/e$ is a solution, it is the solution, i.e., $P = (1/e, 1/e)$. Hence $(dy/dx)_P = (-e)e^{-e(1/e)} = -1$.

So now the gradient of Γ , at the point $P \in \Gamma \cap G$ changes from -1 to 1 as a varies from e^{-e} to $e^{1/e}$. For $a > e^{1/e}$, $(dy/dx)_P > 1$ while for $a < e^{-e}$, $(dy/dx)_P < -1$ and as $a \rightarrow 0$, $(dy/dx)_P \rightarrow -\infty$ and so in the range $0 < a < e^{-e}$, $|(dy/dx)_P| > 1$ and $\Gamma \cap \Gamma^*$ consists of 3 points, as shown in Figure 7.11. In case this graphical consideration does not satisfy the reader, we may prove the above analytically as follows: Since $y = a^x$, $dy/dx = a^x \log a$. Then $(dy/dx)_P = x \log a = \log a^x = \log x$ (since $a^x = y = x$ at P) and this is less than -1 if and only if

$$\begin{aligned}
 \log x < -1 &\Leftrightarrow x < e^{-1} & (*) \\
 &\Leftrightarrow a^x < e^{-1} & (\text{as } x = a^x) \\
 &\Leftrightarrow x \log a < -1 \\
 &\Leftrightarrow \log a < -1/x \\
 &\Leftrightarrow \log a < -e & (\text{since } -1/x < -e \text{ by } (*)) \\
 &\Leftrightarrow a < e^{-e},
 \end{aligned}$$

as required. Now let us collect all the information that we have observed so far that is relevant to the behaviour of $\Gamma : y = a^x$.

Let $a > 0$. Mark off, on the real line, certain points as shown below:

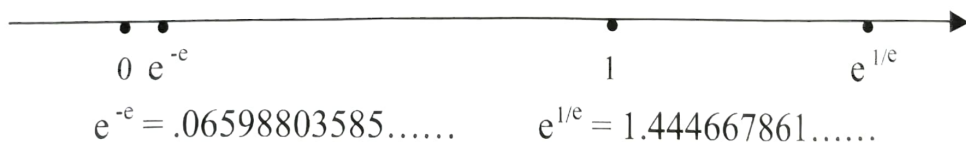


Figure 7.12

Let

$$(i) \Gamma : y = a^x \quad (ii) \Gamma^* : x = a^y \quad (iii) G : y = x,$$

so that Γ, Γ^* are mirror images of each other in G . Then the following results hold:

1. If $e^{-e} \leq a \leq 1$, $\Gamma \cap \Gamma^*$ consists of only one point, say $P = (\alpha, \alpha) \in G$. The gradient of (i) at P is greater than or equal to -1 (it is equal to -1 when $a = e^{-e}$ and is 0 when $a = 1$).

2. If $0 < a < e^{-e}$, $\Gamma \cap \Gamma^*$ consists of 3 points, say, $R = (\alpha, \beta)$, $P = (\gamma, \gamma)$, $Q = (\beta, \alpha)$, $\alpha < \beta$, and the gradient of (i) at P is less than or equal to -1 and as $a \rightarrow 0$, this gradient tends to $-\infty$.
3. If $1 < a < e^{1/e}$, $\Gamma \cap \Gamma^*$ consists of 2 points, say, $P = (\alpha, \alpha)$, $Q = (\beta, \beta)$, $\alpha < \beta$ and the gradient of (i) at P lies between 0 and 1 (it is 0 if $a = 1$ and is 1 if $a = e^{1/e}$; for $a = 1$, Q does not exist, while for $a = e^{1/e}$, Q and P coincide).
4. If $a = e^{1/e}$, $\Gamma \cap \Gamma^*$ consists of one point, say, $P = (\alpha, \alpha) \in G$, in fact G is tangent to both of Γ and Γ^* , and the gradient of (i) at P is 1.
5. If $e^{1/e} < a$, $\Gamma \cap \Gamma^* = \emptyset$.

For the convergence of $\zeta_n = a^{\zeta_{n-1}}$, $\zeta_1 = a$, for $0 < a < 1$, two cases arise:

- (i) $0 < a < e^{-e}$,
- (ii) $e^{-e} \leq a < 1$.

It will be found that in case (ii), ζ_n is convergent to the unique point $P = \Gamma \cap \Gamma^*$ while in case (i), ζ_n has no limit but two limit points (accumulation points) R and Q , as described above and indeed $\zeta_{2n} \rightarrow R$, $\zeta_{2n-1} \rightarrow Q$, as $n \rightarrow \infty$. In fact we have $R \rightarrow (0, 1)$, $Q \rightarrow (1, 0)$, as $a \rightarrow 0$.

Case 5: $e^{-e} \leq a < 1$. Let us take a typical value of a , say $a = 1/2$ ($e^{-e} \leq 1/2 < 1$). The graph of $y = (1/2)^x$ is shown in Figure 7.13.

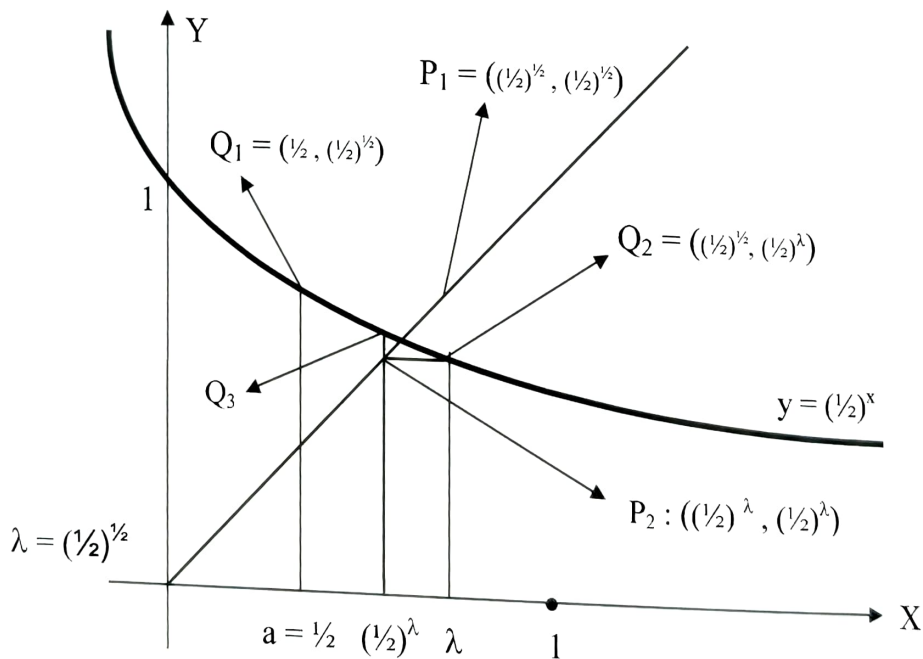


Figure 7.13

The sequence ζ_n , $n = 1, 2, \dots$, is the x -coordinate of Q_n and these Q_n 's are given by a spiral instead of a zigzag as was the case for $1 < a \leq e^{1/e}$. It is now

clear, but perhaps not as clear as in the zigzag case, that the spiral of points Q_i, P_i tends to the unique point of intersection P of Γ and G .

Case 6: $0 < a < e^{-e}$. The relevant graph for a typical a with $0 < a < e^{-e}$, now looks like Figure 7.14. Here, the spirals alternatively tend to the points R and Q , as required.

Analytic proof of Theorem 7.2.1. We now give an analytic proof of all these cases discussed above. More specifically, we prove:

Let $a > 0$ be a real number and let $\zeta_1 = a, \zeta_2 = a^{\zeta_1} = a^a, \dots, \zeta_n = a^{\zeta_{n-1}}$.

- The sequence $\{\zeta_n\}$ is convergent to a limit Λ if and only if $e^{-e} \leq a \leq e^{1/e}$.
- If $1 < a < e^{1/e}$, the equation $x = a^x$ has two roots α, β ($\alpha < \beta$, say) and Λ equals the smaller root α .
- If $a = e^{1/e}$, the equation $x = a^x$ has a unique root α and $\Lambda = \alpha$, while if $a = 1$, trivially $\Lambda = 1$.
- If $e^{1/e} < a$, ζ_n diverges to ∞ .
- If $0 < a < 1$, the ζ_i satisfy

$$0 < \zeta_1 < \zeta_3 < \zeta_5 < \dots < \zeta_6 < \zeta_4 < \zeta_2 < 1.$$

Writing $A = \lim \zeta_{2n-1}, B = \lim \zeta_{2n}$, we have that if $e^{-e} \leq a \leq 1, A = B = \Lambda$. If $0 < a < e^{-e}$, then $A < B$ and ζ_n oscillates with A, B as two limit points.

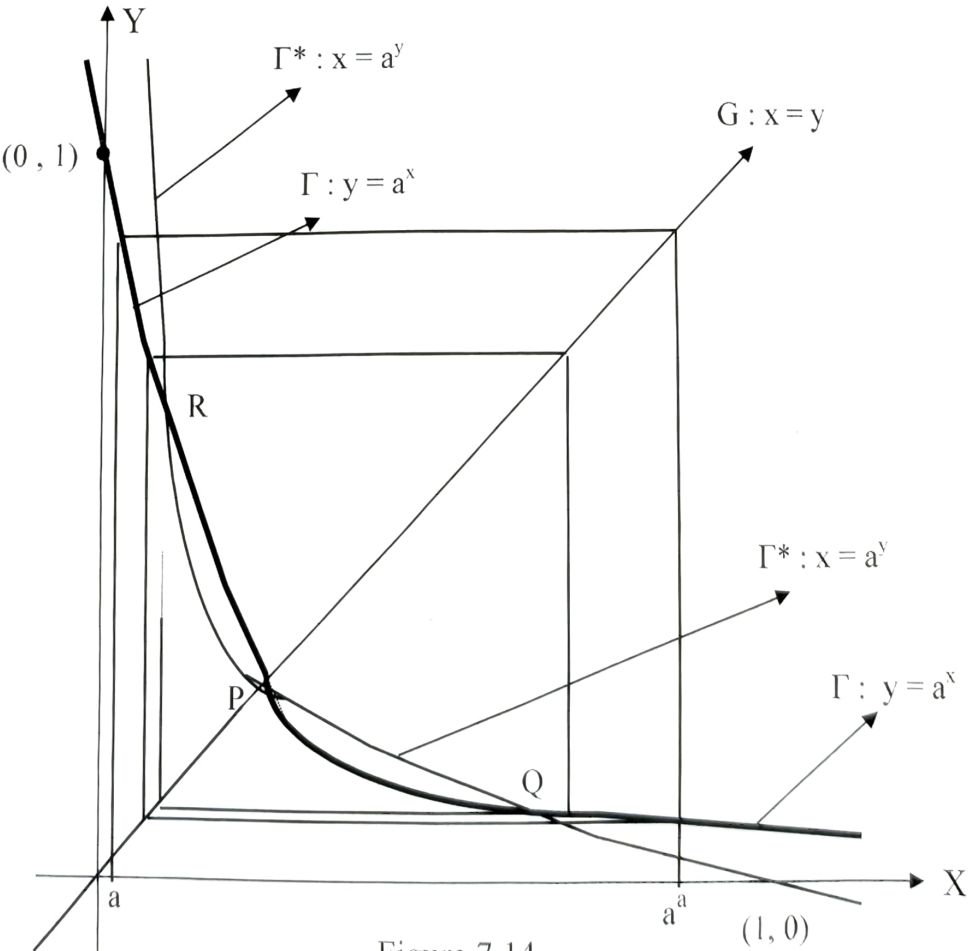


Figure 7.14

First, let $1 < a \leq e^{1/e}$. We have $\zeta_n/\zeta_{n-1} = a^{\zeta_{n-1}}/a^{\zeta_{n-2}} = a^{(\zeta_{n-1}-\zeta_{n-2})}$. Hence $\zeta_n > \zeta_{n-1}$ if and only if $\zeta_{n-1} > \zeta_{n-2}$. But $\zeta_2 = a^a > a = \zeta_1$, so by induction $\zeta_n > \zeta_{n-1}$ for all n , i.e. the sequence $\{\zeta_n\}$ is increasing.

Next we show that the sequence $\{\zeta_n\}$ is bounded above by e . First notice that, since $a \leq e^{1/e} < e$, so $a^a \leq (e^{1/e})^a < (e^{1/e})^e = e$ (since $a < e$). Now use induction. If $\zeta_{n-1} < e$, then $\zeta_n = a^{\zeta_{n-1}} \leq (e^{1/e})^{\zeta_{n-1}} < (e^{1/e})^e = e$ (by induction hypothesis). It follows that ζ_n tends to a limit, Λ say, which is less than or equal to e . Letting $n \rightarrow \infty$ in the defining equation $\zeta_n = a^{\zeta_{n-1}}$, we get $\Lambda = a^\Lambda$, where $\Lambda \leq e$, i.e.,

Λ is a root of the equation $x = a^x$ and $\Lambda \leq e$.

(#)

To get this root, we draw the graphs $\Gamma : y = a^x$ and $G : y = x$. In the range $1 < a \leq e^{1/e}$, the graphs Γ and G , for $a = 1$, $a = e^{1/e}$ and a typical a are shown in Figure 7.15.

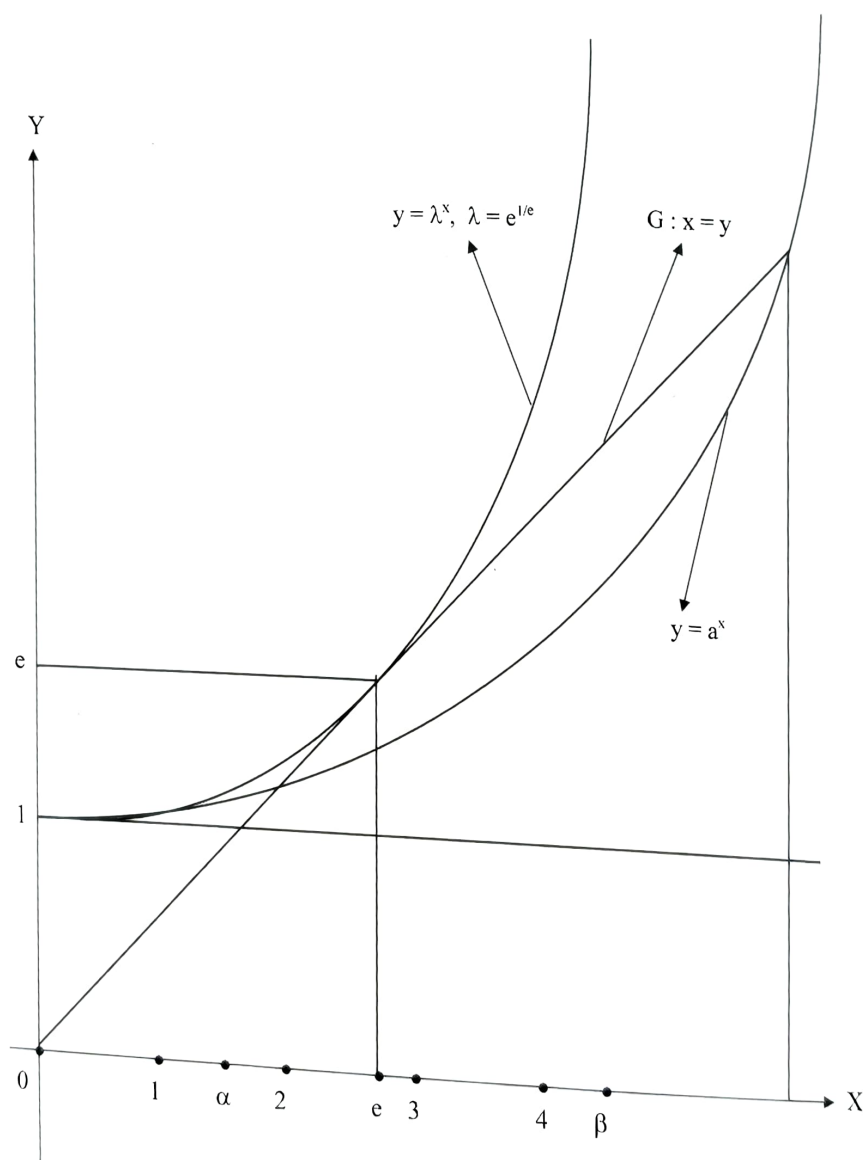


Figure 7.15

For $a \neq e^{1/e}$, the equation $x = a^x$ has two roots α, β ($\alpha < \beta$, say) and indeed $\alpha < e < \beta$. Also, since $\Lambda \leq e$, it follows that $\Lambda = \alpha$, the smaller of the two roots.

Next, let $e^{1/e} < a$. We have seen that in this case, the two curves Γ and G have no real intersection (see case 3 and Figure 7.8 of the graphical motivation). Suppose $\lim \zeta_n$ exists and equals Λ . Then

$$0 = a^\Lambda - a^\Lambda = \lim(a^{\zeta_n} - a^\Lambda) = \lim(\zeta_{n+1} - a^\Lambda) = \Lambda - a^\Lambda,$$

i.e., that Λ is a root of the equation $x = a^x$, showing that $\Gamma \cap G \neq \emptyset$.

Let now $0 < a < 1$. We show, by induction, that

$$0 < \zeta_1 < \zeta_3 < \zeta_5 < \dots < \zeta_6 < \zeta_4 < \zeta_2 < 1.$$

We need one pertinent inequality throughout, viz., if $0 < a < 1$ (as is the case) and if $0 < p < q$, then

$$0 < a^q < a^p < 1. \tag{*}$$

The proof of this is immediate from the graph Γ of $y = a^x$ ($0 < a < 1$), which has the shape shown in Figure 7.16.

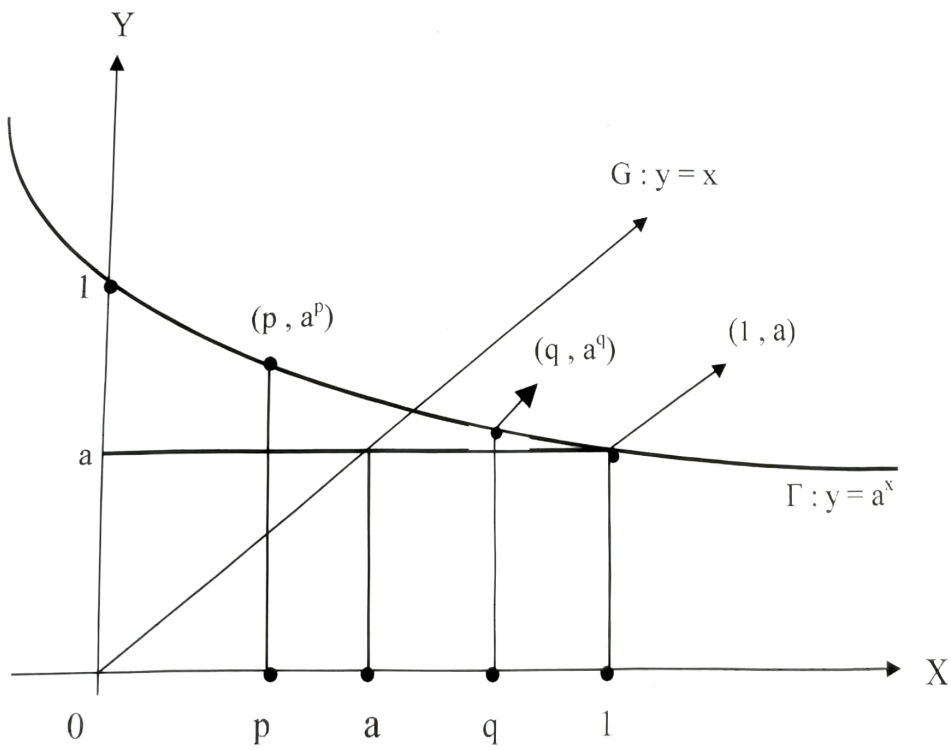


Figure 7.16

Since, as x increases, Γ decreases (i.e., comes down onto the x -axis), it follows that $a^q < a^p$ and both these numbers are less than 1 (even if q is to the right of 1).

Now using this inequality repeatedly, we have

$$\begin{aligned}
 0 < \zeta_1 &= a^1 < a^a = \zeta_2 < 1, \\
 \zeta_3 &= a^{(a^a)} < a^a = \zeta_2, & (\text{since } a < a^a) \\
 \zeta_1 &= a^1 < a^{(a^a)} = \zeta_3, & (\text{since } a^a < 1) \\
 \zeta_2 &= a^a > a^{(a^{(a^a)})} = \zeta_4,
 \end{aligned}$$

since $a < a^{(a^a)}$ and since $\zeta_3 < \zeta_2$, so $a^{\zeta_2} < a^{\zeta_3}$, i.e., $\zeta_3 < \zeta_4$. These inequalities imply $\zeta_1 < \zeta_3 < \zeta_4 < \zeta_2$, which starts off our induction.

Now the induction step. Suppose (induction hypothesis)

$$\zeta_1 < \zeta_3 < \dots < \zeta_{2r-1} < \zeta_{2r} < \dots < \zeta_4 < \zeta_2.$$

Then

$$\begin{aligned}
 \zeta_{2r+1} &= a^{\zeta_{2r}} < a^{\zeta_{2r-1}} = \zeta_{2r}, & (\text{since } \zeta_{2r-1} < \zeta_{2r}) \\
 \zeta_{2r-1} &= a^{\zeta_{2r-2}} < a^{\zeta_{2r}} = \zeta_{2r+1}, & (\text{since } \zeta_{2r} < \zeta_{2r-2}) \\
 \zeta_{2r+2} &= a^{\zeta_{2r+1}} < a^{\zeta_{2r-1}} = \zeta_{2r}, & (\text{since } \zeta_{2r-1} < \zeta_{2r+1}) \\
 \zeta_{2r+1} &= a^{\zeta_{2r}} < a^{\zeta_{2r+1}} = \zeta_{2r+2}, & (\text{since } \zeta_{2r+1} < \zeta_{2r})
 \end{aligned}$$

i.e.,

$$\zeta_1 < \zeta_3 < \dots < \zeta_{2r-1} < \zeta_{2r+1} < \zeta_{2r+2} < \zeta_{2r} < \zeta_{2r-2} \dots < \zeta_4 < \zeta_2,$$

which gives the induction step.

Finally, to see that $0 < \zeta_n < 1$ for all n , we proceed as follows:

Since $0 < a < 1$, so $0 < a^a < 1$, i.e., $0 < \zeta_2 < 1$. If $0 < \zeta_{n-1} < 1$, then $0 < a^{\zeta_{n-1}} < 1$, i.e., $0 < \zeta_n < 1$, so by induction, the result follows.

Thus ζ_{2n-1} increases while ζ_{2n} decreases and both are bounded by 0 and 1. Hence they tend to limits, say ζ_{2n-1} tends to A , ζ_{2n} tends to B , where

$$0 < A \leq B \leq 1.$$

Now, in the relation $\zeta_n = a^{\zeta_{n-1}}$, let n tend to ∞ , first through odd values and then through even values and we get $A = a^B$, $B = a^A$, i.e., (A, B) is a point common to the two curves $\Gamma : y = a^x$ and $\Gamma^* : x = a^y$. But we have seen that if $e^{-e} \leq a \leq 1$, then $\Gamma \cap \Gamma^*$ is a single point P , which belongs to $G : y = x$. It follows that $A = B = \lim \zeta_n$ exists (in case $e^{-e} \leq a \leq 1$) and is the unique solution of the equation $x = a^x$ (i.e., the unique intersection of Γ , Γ^* and G), as required.

Let $0 < a < e^{-e}$. In this case $\Gamma \cap \Gamma^*$ consists of three distinct points $R = (\alpha, \beta)$, $P = (\gamma, \gamma)$, $Q = (\beta, \alpha)$, ($\alpha < \beta$). Since $(A, B) \in \Gamma \cap \Gamma^*$, we see that if $A \neq B$, then $A = \alpha$, $B = \beta$ and although $\{\zeta_n\}$ has two limit points, it has no limit. It remains therefore to show that $A \neq B$.

The equations $A = a^B$, $B = a^A$ (established above) imply $A = a^B = a^{(a^A)}$ and $B = a^A = a^{(a^B)}$, i.e., that A, B are roots of the equation $x = a^{(a^x)}$. Let

$$\phi(x) = x - a^{(a^x)} . \tag{1}$$

If Λ is a root of ϕ , i.e., $\Lambda = a^{(a^\Lambda)}$, then $\log \Lambda = a^\Lambda \log a$ or $\log(1/\Lambda) = a^\Lambda \log(1/a)$ and so $\log \log(1/\Lambda) = \Lambda \log a + \log \log(1/a)$. Let

$$\psi(x) = \log \log(1/a) + x \log a - \log \log(1/x) . \tag{2}$$

Then it is easy to check that $\phi(x) = 0$ if and only if $\psi(x) = 0$, and indeed

$$\begin{array}{ccc} & > & \\ \phi(x) \text{ is } & = 0 & \Leftrightarrow \psi(x) \text{ is } = 0 \\ & < & \end{array} \tag{3}$$

(for example, $\phi(x_0) > 0 \Leftrightarrow x_0 > a^{(a^{x_0})} \Leftrightarrow \log x_0 > a^{x_0} \log a \Leftrightarrow \log(1/x_0) < -a^{x_0} \cdot \log a \Leftrightarrow \log \log(1/x_0) < x_0 \log a + \log(-\log a) \Leftrightarrow \log \log(1/x_0) < x_0 \log a + \log \log(1/a) \Leftrightarrow \psi(x_0) > 0$). Also

$$\psi'(x) = \log a - (1/\log(1/x)) \cdot (1/(1/x)) \cdot (-1/x^2) = \log a - 1/(x \log x) . \tag{4}$$

So $\psi'(x) = 0$ when $\log x^x = 1/\log a$, i.e., when $x^x = e^{1/\log a}$.

We now consider the equation $x^x = c$ ($0 < c < 1$) and record:

Lemma 7.2.2. *The equation $x^x = c$ ($0 < c < 1$) has no real root if $0 < c < 1/e^{1/e}$; one real root if $c = 1/e^{1/e}$; two real roots if $1/e^{1/e} < c < 1$.*

Proof. Consider the graph T of $y = x^x$ (sketch shown in Figure 7.17). Taking logarithms and differentiating gives $(1/y) \cdot dy/dx = 1 + \log x$; so there is an extreme at $x = 1/e$ and indeed it is a minimum, equal to $e^{-1/e}$. As x increases from 0 to 1, $y = x^x$ decreases from 1 to the value $1/e^{1/e}$ at $x = 1/e$ and then increases from $1/e^{1/e}$ to 1.

To continue with the analytic proof, as $\psi'(x) = 0$ when $x^x = e^{1/\log a}$ ($= c$, say), we get that $\psi'(x) = 0$ has no root, one root and two roots according as $0 < e^{1/\log a} < e^{-1/e}$, $e^{1/\log a} = e^{-1/e}$ and $e^{-1/e} < e^{1/\log a} < 1$ respectively, i.e., according as $a > e^{-e}$, $a = e^{-e}$ and $a < e^{-e}$, respectively (check that $1/\log a$ is less than, equal to or greater than $-1/e \Leftrightarrow \log a$ is less than, equal to or greater than $-e \Leftrightarrow a$ is less than, equal to or greater than e^{-e} respectively).

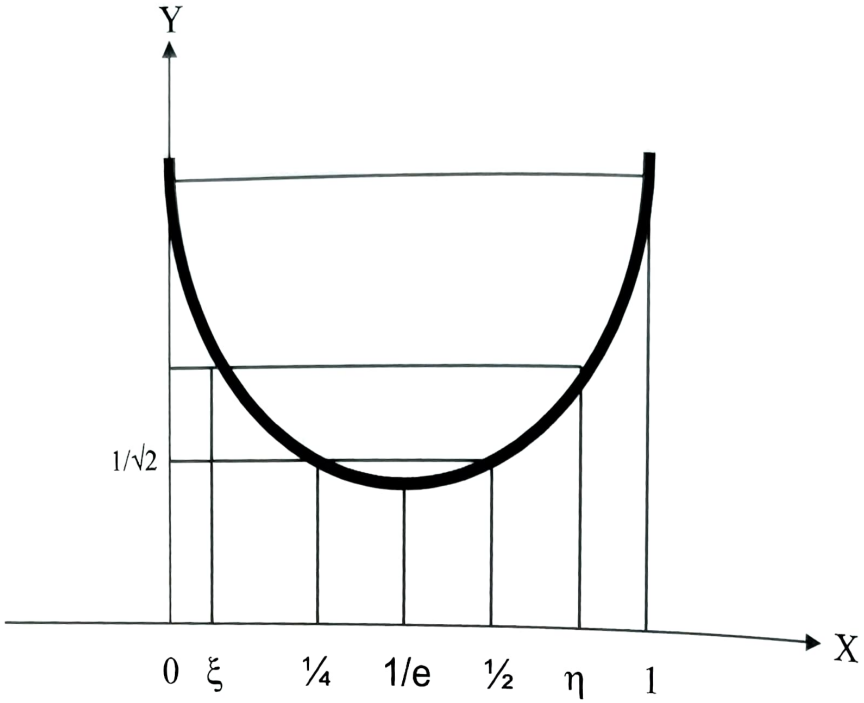


Figure 7.17

Now consider the graph of $y = \psi(x)$ and note that as $x \rightarrow 0$, $\psi(x) \rightarrow -\infty$ and as $x \rightarrow 1$, $\psi(x) \rightarrow \infty$. It follows that $\psi(x)$ crosses the x -axis an odd number of times (see Figures 7.18–7.21), i.e., that $\psi(x) = 0$ has an odd number of roots. We now split two cases.

Case (i): $e^{-e} \leq a < 1$. Then by above $\psi'(x) = 0$ has either no root or one root and so only the two possibilities (i), (i)' in the diagrams (Figures 7.18–7.21) hold, since in the other possibilities, the equation $\psi'(x) = 0$ has 2, 4, ... roots respectively.

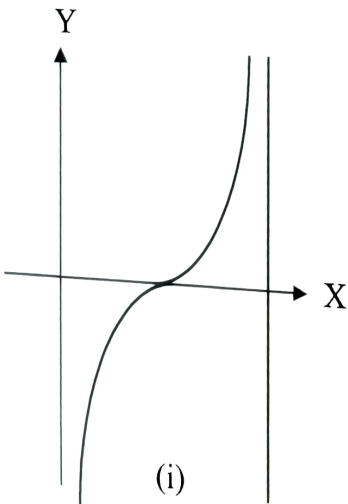


Figure 7.18

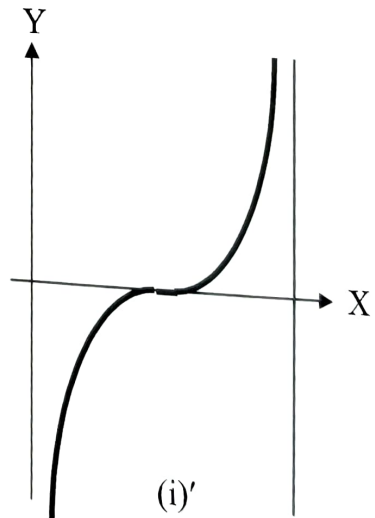


Figure 7.19

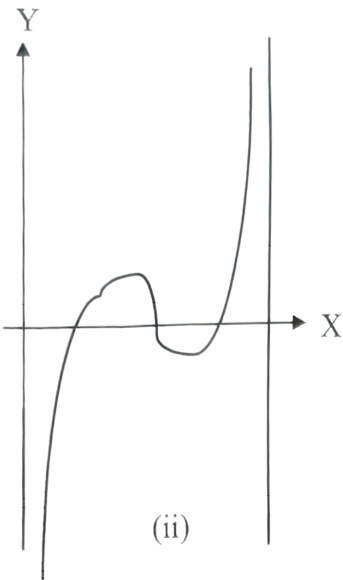


Figure 7.20

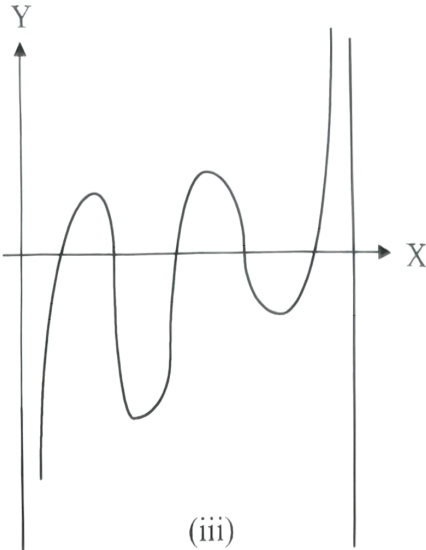


Figure 7.21

Thus $\psi(x) = 0$ has only one root and hence $\phi(x) = 0$ also has only one root (see (3)). Since A, B are roots of $\phi(x) = 0$, it follows that $A = B$; so in this case the sequence ζ_n converge to $A = B$.

Case (ii): $0 < a < e^{-e}$. We have seen in this case that the equation $x^x = e^{1/\log a} (= c)$ has two roots x_1, x_2 , say, in the interval $(0, 1)$, i.e.,

$$x_2^{x_2} = x_1^{x_1} = e^{1/\log a} \tag{5}$$

(for we must check that $c = e^{1/\log a} > 1/e^{1/e}$, i.e., that $1/\log a > -1/e$, i.e., that $\log a < -e$, i.e., that $a < e^{-e}$, which is true in this case).

We have also seen in Lemma 7.2.2 that these two roots lie on different sides of $1/e$, i.e.,

$$x_1 < 1/e < x_2 . \tag{6}$$

Then $x < x_1$ implies that $x^x > x_1^{x_1} = e^{1/\log a} \Rightarrow \psi'(x) > 0$ (see Figure 7.22).

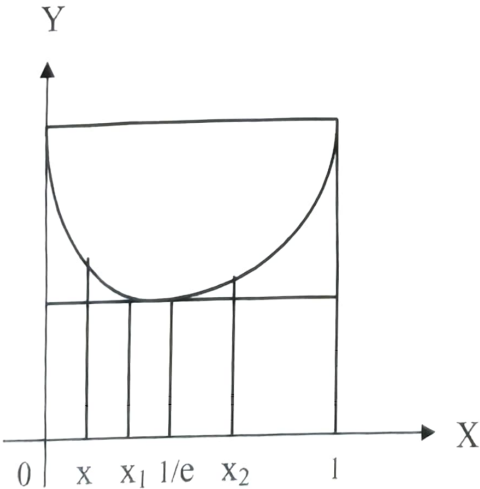


Figure 7.22

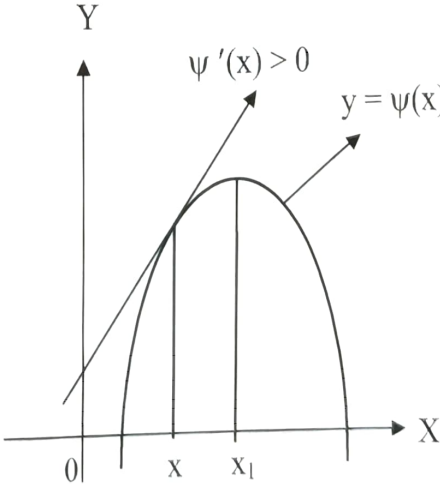


Figure 7.23

To check the above implications; the first one follows by looking at the graph of the function $y = x^x$ in $[0, 1]$, since $x < x_1$. To see the second implication, we have $\psi'(x) > 0 \Leftrightarrow \log a > 1/\log(x^x)$ (see (4)) $\Leftrightarrow 1/\log a < \log(x^x) \Leftrightarrow x^x > e^{1/\log a}$, as required.

Similarly, $x > x_1 \Rightarrow x^x < e^{1/\log a} \Rightarrow \psi'(x) < 0$. So (see Figure 7.23)

$x = x_1$ is a maximum of $\psi(x)$ & similarly $x = x_2$ is a minimum of $\psi(x)$. (7)

In particular,

$$\psi'(x_1) = \psi'(x_2) = 0. \quad (8)$$

So now, by (2), we have $\psi(x_1) = \log \log(1/a) + x_1 \log a - \log \log(1/x_1)$. However, by (8), $\psi'(x_1) = 0$ and so by (4), $\log a = (1/x_1) \log x_1$, so $x_1 \log a = 1/\log x_1 = -1/\log(1/x_1)$. Therefore, we have

$$\begin{aligned} \log \log(1/a) &= \log 1 - \log(-x_1 \log x_1) \\ &= \log(1/(-x_1 \log x_1)) \\ &= \log(1/x_1 \log(1/x_1)) \\ &= \log((1/x_1)/\log(1/x_1)) \\ &= \log(1/x_1) - \log \log(1/x_1). \end{aligned}$$

So

$$\begin{aligned} \psi(x_1) &= (\log(1/x_1) - \log \log(1/x_1)) - 1/(\log(1/x_1)) - \log \log(1/x_1) \\ &= r - 1/r - 2 \log r = \lambda(r), \text{ say,} \end{aligned} \quad (9)$$

where

$$r = \log(1/x_1) > \log e = 1 \quad (\text{since } x_1 < 1/e). \quad (10)$$

Then $\lambda'(r) = 1 + 1/r^2 - 2/r = (r-1)^2/r^2 \geq 0$ and in fact it is greater than 0, since $r > 1$, by (10), i.e., $\lambda'(r) > 0$, showing that $\lambda(r)$ is increasing and so $\lambda(r) > \lambda(1) = 0$, or by (9) we have that

$$\psi(x_1) > 0. \quad (11)$$

Further, by (2)

$$\psi(1/e) = \log \log(1/a) + (1/e) \log a = \log \log(1/a) - (1/e) \log(1/a). \quad (12)$$

But

$$(\log z)/z < 1/e, \text{ if } z > e \quad (13)$$

(check: We have to show that $h(z) = z - e \log z > 0$. We have $dh/dz = 1 - e/z > 0$ if $z > e$, and so $h(x)$ is increasing and $h(e) = 0$, so $h(z) > 0$).

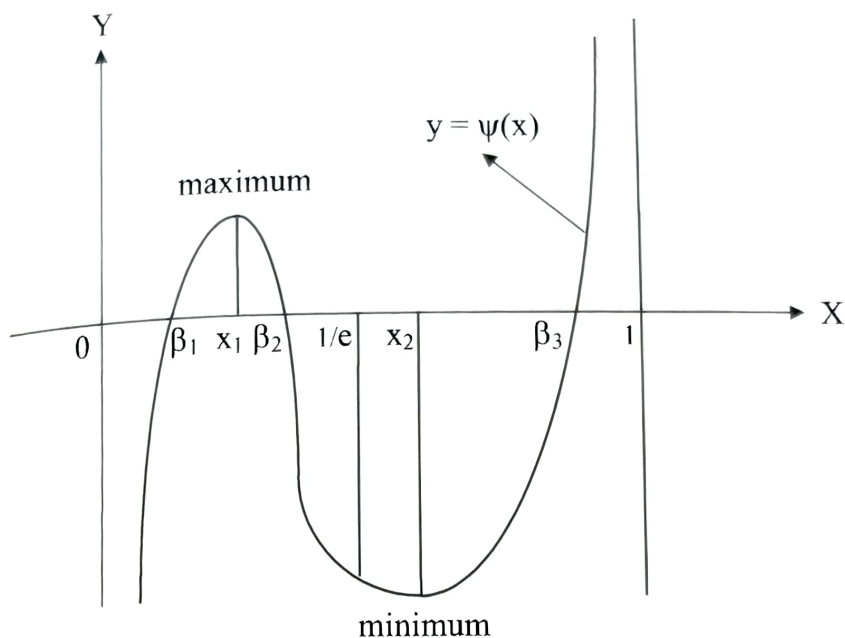


Figure 7.24

Taking $z = \log(1/a)$, this shows that $(\log \log(1/a))/\log(1/a) < 1/e$ (for, $z = \log(1/a)$ is greater than e since $1/a > e^e$ as $a < e^{-e}$), i.e., $\log \log(1/a) < (1/e)\log(1/a)$ (since $\log(1/a) > 0$ as $a < 1$), and so, by (12),

$$\psi(1/e) < 0. \quad (14)$$

It follows that the graph of $\psi(x) = 0$ is as shown in Figure 7.24 (see (6), (7) and (11)), showing that $\psi(x) = 0$ has three roots in $(0, 1)$; call them $\beta_1, \beta_2, \beta_3$, where

$$0 < \beta_1 < x_1 < \beta_2 < 1/e < x_2 < \beta_3 < 1.$$

Since, by (3), $\phi(x)$ is larger than, equal to or less than 0 if and only if $\psi(x)$ is larger than, equal to or less than 0, respectively, it follows that $\phi(x)$ has the same three roots $\beta_1, \beta_2, \beta_3$. Remembering again that $0 < a < e^{-e}$, we now show that $B = \beta_3$, $A = \beta_1$. Indeed, we shall show, by induction, that $\zeta_{2n} > 1/e$ for all n . We have $\zeta_0 = 1 > 1/e$. If $\zeta_{2n-2} > 1/e$, then $\zeta_{2n} = a^{a^{\zeta_{2n-2}}} > a^{a^{1/e}} > 1/e$ (this last inequality, since $\psi(1/e) < 0$ by (14) and so $\phi(1/e) < 0$). Thus, $\zeta_{2n} > 1/e$ and so $\lim \zeta_{2n} = B \geq 1/e$. But B is a zero of ϕ and the only zero of ϕ larger than $1/e$ is β_3 . So $B = \beta_3$.

Next we show, again by induction, that $\zeta_{2n+1} < x_1$ for all n . The induction step is simple. Suppose $\zeta_{2n-1} < x_1$, then $\zeta_{2n+1} = a^{a^{\zeta_{2n-1}}} < a^{a^{x_1}} < x_1$ (the last inequality comes from (11) because $\psi(x_1) > 0$ and so $\phi(x_1) > 0$), completing the induction step.

To start the induction, it remains to show that $a < x_1$. Now $z \geq e \log z$ for all $z > 0$ (check: It is enough to prove that $g(z) = z - e \log z \leq 0$. Now

$g'(z) = 1 - e/z = 0$ if $z = e$ and $g''(z) = e/z^2 > 0$ at $z = e$, and so $z = e$ is a minimum and $g(e) = 0$, so $g(z) \leq 0$. Hence $1/\sqrt{x_1} \geq e \log 1/\sqrt{x_1} > 2 \log 1/\sqrt{x_1} = \log 1/x_1$, i.e., $1/x_1 > (\log 1/x_1)^2$, i.e., $1/x_1 \log(1/x_1) > \log 1/x_1$ or $1/(-x_1 \log 1/x_1) > -\log x_1$. But $\log a = 1/(x_1 \log x_1) < \log x_1$ (taking logarithms in (5)), by above, we get that $a < x_1$, as required.

So now, $\zeta_{2n+1} < x_1$ for all n and ζ_{2n+1} is increasing and tends to A , so $A \leq x_1$. But A is a zero of ϕ and the only zero of ϕ less than or equal to x_1 is β_1 . Hence $A = \beta_1$. Thus $A \neq B$ and so ζ_n oscillates with A, B as limits.

This at last completes the analytic proof of Theorem 7.2.1.

§7.3. Applications and examples

In the last section we have given graphical and analytic proofs of the convergence or otherwise of the infinite exponential of the title of this chapter. Our object now is to establish the connection of this infinite exponential with the function $f(x, y) = x^y - y^x$ in the range of convergence of the exponential. We have the following striking result:

Theorem 7.3.1. *In its interval of convergence, i.e., for $z \in [e^{-e}, e^{1/e}]$, the functions*

$$x = h(z) = z^{z^{z^{\cdot^{\cdot^{\cdot}}}}} \quad \text{and} \quad z = g(x) = x^{1/x}$$

are partial inverses of each other. More specifically

(i) $g(h(z)) = z$, for all $z \in [e^{-e}, e^{1/e}]$ (= the full range of convergence of $h(z)$),

(ii) $h(g(x)) = x$, for all $x \in [1/e, e]$.

Proof. Since exponentiation is continuous (and increasing), it follows that

$$x = h(z) \Rightarrow z^x = z^{h(z)} \Rightarrow z^x = z^{(z^{z^{z^{\cdot^{\cdot^{\cdot}}}}})} = h(z) = x \Rightarrow z = x^{1/x} = g(x).$$

Hence, we have

(i) $g(h(z)) = g(z^{z^{z^{\cdot^{\cdot^{\cdot}}}}}) = g(x) = z$; and

(ii) $h(g(x)) = h(x^{1/x}) = h(z) = x$.